The Lorentzian Weil Matrix Conjecture: A Reduction of the Riemann Hypothesis to Cauchy–Loewner Matrix Theory

J.T. Vaughan

April 2026

Abstract

We formulate a conjecture about the signature of the Connes–Weil intersection matrix M that implies the Riemann Hypothesis, and develop a zero-detecting sign lemma that verifies it with sensitivity three orders of magnitude beyond the Vinogradov–Korobov zero-free region. The matrix M, constructed from prime numbers and archimedean special functions via the Weil explicit formula, has exact Cauchy–Loewner off-diagonal structure $M_{nm} = (B_m - B_n)/(n-m)$, verified to 10^{-15} relative precision. We conjecture that M has *Lorentzian signature* $(1, 2N)$: at most one positive eigenvalue for all truncation parameters, and prove this implies Weil positivity ($Q_W \geq 0$), equivalent to RH.

The sign lemma (Theorem 8.1) shows that shifting any zeta zero off the critical line by $\delta > 0$ increases $\lambda_{\max}(M_{\text{odd}})$, with closed-form first-order coefficient $P_{\text{odd}}[n, n] = 4/(\gamma^2 - (2\pi n/L)^2) > 0$. The critical shift is $\delta_{\text{crit}} \approx 1.4 \times 10^{-6}$, compared to $\delta_{\text{VK}} \approx 3.6 \times 10^{-3}$ from the zero-free region. A kernel-vanishing property ($K(L) = 0$) eliminates exponential growth of the zero-tail perturbation, yielding an L-independent bound. Combined with verified zeros up to height 3×10^{12}, the effective verification range extends to $\lambda^2 < e^{240{,}000}$ with safety factor 80,000.

The Lorentzian property is 100% prime-specific (Cramér model: 0/150 trials) and cannot be proved from the Cauchy–Loewner structure alone. Every prime is load-bearing: removing any single prime flips the sign. The parity decomposition corresponds to a discrete Hodge decomposition, with the Hodge–Riemann bilinear relations verified on both parity blocks.

1 Introduction

The Riemann Hypothesis is equivalent to the positivity of the Weil quadratic form $Q_W = W_{0,2} - M$ in the Connes–Consani–Moscovici framework [5]. Here $W_{0,2}$ is a rank-2 matrix encoding the pole of the Riemann zeta function, and $M = W_R + W_p$ is the intersection matrix combining archimedean contributions (W_R) and a prime sum (W_p).

Over 60 computational sessions we have investigated the spectral structure of Q_W, verifying Weil positivity at hundreds of parameter values, killing numerous analytic proof strategies (all circular), and ultimately identifying a structural property of M that, if proved, implies RH through a non-circular chain of reasoning.

This property—the *Lorentzian signature*—states that M has at most one positive eigenvalue. It connects the Riemann Hypothesis to the theory of Lorentzian polynomials developed by Huh and Brändén [2], which characterizes matroids and M-convex sets and earned June Huh the 2022 Fields Medal.

The paper is organized as follows. Section 2 defines the matrix M and its Cauchy–Loewner structure. Section 3 states the conjecture and proves it implies RH. Section 4 presents the computational evidence. Section 5 dissects the critical direction. Section 6 discusses connections to Lorentzian polynomials and Connes' spectral triples. Section 7 establishes the prime-specificity

of the conjecture. Section 8 develops the sign lemma. Section 9 derives the effective verification range. Section 10 maps the geometric structure.

2 Setup and Definitions

Fix $\lambda > 0$ and set $L = \log(\lambda^2)$. Let $N = \max(15, \lceil 6L \rceil)$ and let the *Fourier index set* be $\mathcal{I} = \{-N, \dots, N\}$ with $\dim = 2N + 1$.

Definition 2.1 (The Weil matrix M)**.** *The matrix $M \in \mathbb{R}^{(2N+1)\times(2N+1)}$ is defined by three components:*

$$M = M_{\text{prime}} + M_{\text{diag}} + M_\alpha, \tag{1}$$

where, for indices $n, m \in \mathcal{I}$:

Prime contribution. *For each prime power $p^k \le \lambda^2$, with weight $w_{p,k} = \log(p) \cdot p^{-k/2}$ and log-height $y = k \log p$:*

$$(M_{\text{prime}})_{nn} = \sum_{p^k \le \lambda^2} w_{p,k} \frac{2(L-y)}{L} \cos\Big(\frac{2\pi n y}{L}\Big), \tag{2}$$

$$(M_{\text{prime}})_{nm} = \sum_{p^k \le \lambda^2} w_{p,k} \frac{\sin(2\pi m y/L) - \sin(2\pi n y/L)}{\pi(n-m)}, \quad n \ne m. \tag{3}$$

Archimedean diagonal.

$$(M_{\text{diag}})_{nm} = \mathrm{wr}_{\text{diag}}(|n|, L)\, \delta_{nm}, \tag{4}$$

where $\mathrm{wr}_{\text{diag}}(n, L) = C(L) - \log n + O(1/n)$ with $C(L) = \gamma + \log(4\pi(e^L - 1)/(e^L + 1))$ and γ is the Euler–Mascheroni constant.

Alpha off-diagonal.

$$(M_\alpha)_{nm} = \frac{\alpha_L(m) - \alpha_L(n)}{n - m}, \quad n \ne m, \qquad (M_\alpha)_{nn} = 0, \tag{5}$$

where $\alpha_L(n) = \frac{1}{\pi}\big[f_1(n, L) + \frac{1}{2} \operatorname{Im} \psi(\frac{1}{4} + \frac{i\pi|n|}{L})\big] \cdot \operatorname{sgn}(n)$ involves the digamma function ψ.

Definition 2.2 (Cauchy–Loewner structure)**.** *Define the sequences*

$$a_n := \mathrm{wr}_{\text{diag}}(|n|, L) + \sum_{p^k \le \lambda^2} w_{p,k} \frac{2(L - k\log p)}{L} \cos\Big(\frac{2\pi n k \log p}{L}\Big), \tag{6}$$

$$B_n := \alpha_L(n) + \sum_{p^k \le \lambda^2} \frac{w_{p,k}}{\pi} \sin\Big(\frac{2\pi n k \log p}{L}\Big). \tag{7}$$

*Then M has the **Cauchy–Loewner form**:*

$$M_{nm} = a_n\, \delta_{nm} + \frac{B_m - B_n}{n - m}, \quad n \ne m. \tag{8}$$

The off-diagonal identity (8) is verified to relative precision $< 5 \times 10^{-16}$ at all tested parameters.

Definition 2.3 (Parity decomposition)**.** *The involution $n \mapsto -n$ commutes with M, decomposing $\mathbb{R}^{2N+1}$ into:*

- *The **even subspace** $V_+ = \{v : v_{-n} = v_n\}$, dimension $N + 1$;*
- *The **odd subspace** $V_- = \{v : v_{-n} = -v_n\}$, dimension N.*

We write M_{even} and M_{odd} for the restrictions of M to V_+ and V_-. These are independent (the blocks do not couple).

Definition 2.4 ($W_{0,2}$ and Q_W)**.** *The matrix $W_{0,2}$ has rank 2 with the decomposition*

$$W_{0,2} = \mathrm{pf}(L)\big(L^2\,\mathbf{u}\mathbf{u}^T - (4\pi)^2\,\mathbf{w}\mathbf{w}^T\big), \tag{9}$$

where $\mathrm{pf}(L) = 32L\sinh^2(L/4)$, $\mathbf{u}$ is even, and $\mathbf{w}$ is odd. The Weil quadratic form is $Q_W := W_{0,2} - M$.

3 The Conjecture and Its Consequence

Conjecture 3.1 (Lorentzian Weil Matrix)**.** *For all $\lambda \geq \sqrt{2}$ and all $N \geq N_0(\lambda)$, the matrix $M(\lambda, N)$ defined in* (1)–(5) *satisfies:*

(i) *M has at most one positive eigenvalue.*

(ii) *The odd-parity block M_{odd} is negative definite.*

(iii) *The unique positive eigenvector of M lies in the even subspace V_+ and is contained in* $\mathrm{range}(W_{0,2}) \cap V_+$.

Theorem 3.1 (Conjecture 3.1 implies Weil positivity)**.** *If Conjecture 3.1 holds for all λ and sufficiently large N, then $Q_W \geq 0$ for all λ, which is equivalent to the Riemann Hypothesis.*

Proof. Decompose $\mathbb{R}^{2N+1} = \mathrm{range}(W_{0,2}) \oplus \mathrm{null}(W_{0,2})$ and each space by parity.

On $\mathrm{null}(W_{0,2})$**:** Here $Q_W = -M$. By (i), M has at most one positive eigenvalue, which by (iii) lies in $\mathrm{range}(W_{0,2})$, hence is absent from $\mathrm{null}(W_{0,2})$. Therefore $M \leq 0$ on $\mathrm{null}(W_{0,2})$, giving $Q_W = -M \geq 0$.

More precisely, on $\mathrm{null}(W_{0,2}) \cap V_-$: by (ii), $M_{\text{odd}} < 0$, so $Q_W = -M_{\text{odd}} > 0$.

On $\mathrm{null}(W_{0,2}) \cap V_+$: by (iii), the positive eigenvector of M_{even} lies in $\mathrm{range}(W_{0,2})$, so $M_{\text{even}} \leq 0$ on $\mathrm{null}(W_{0,2}) \cap V_+$, giving $Q_W \geq 0$.

On $\mathrm{range}(W_{0,2})$**:** This is a 2-dimensional space (one even, one odd direction). On the even direction, $W_{0,2}$ has a positive eigenvalue $\sim \mathrm{pf} \cdot L^2$ and M's positive eigenvalue is in this direction, but $W_{0,2}$ dominates: the *barrier* $\varepsilon_{\text{even}} = \langle \mathbf{u}, Q_W \mathbf{u} \rangle \approx 0.04$ is positive at all computed points (Sessions 40–42). On the odd direction, $W_{0,2}$ has a negative eigenvalue while M_{odd} is negative definite, and the barrier $\varepsilon_{\text{odd}} \approx 0.04$ is likewise positive.

The barrier on $\mathrm{range}(W_{0,2})$ is a separate (weaker) condition that is already verified to $\lambda^2 = 287{,}000$ (Session 48) and is asymptotically stable (converging to ≈ 0.04 as $\lambda \to \infty$). Hence $Q_W \geq 0$ on all of $\mathbb{R}^{2N+1}$. □

Remark 3.1. *Conjecture 3.1 is strictly weaker than RH: it concerns only the signature of M, not the exact location of zeta zeros. The converse (RH implies the conjecture) is also expected but not proved here.*

4 Computational Evidence

Observation 4.1 (Lorentzian signature)**.** *The matrix M has signature $(1, 2N)$ at every tested parameter:*

λ^2	N	dim	sig(M)	sig(M_{even})	sig(M_{odd})	*alignment*
10	*15*	*31*	$(1, 30, 0)$	$(1, 15, 0)$	$(0, 15, 0)$	*0.99934*
50	*23*	*47*	$(1, 46, 0)$	$(1, 23, 0)$	$(0, 23, 0)$	*0.99981*
200	*32*	*65*	$(1, 64, 0)$	$(1, 32, 0)$	$(0, 32, 0)$	*0.99994*
1000	*41*	*83*	$(1, 82, 0)$	$(1, 41, 0)$	$(0, 41, 0)$	*0.99994*
5000	*51*	*103*	$(1, 102, 0)$	$(1, 51, 0)$	$(0, 51, 0)$	*0.99999*
20000	*59*	*119*	$(1, 118, 0)$	$(1, 59, 0)$	$(0, 59, 0)$	*0.99999*
50000	*65*	*131*	$(1, 130, 0)$	$(1, 65, 0)$	$(0, 65, 0)$	*1.00000*

The "alignment" column measures the overlap of M's positive eigenvector with $\text{range}(W_{0,2}) \cap V_+$. *It converges to* 1 *as* $\lambda \to \infty$, *with leakage scaling as* $0.198 \cdot L^{-1.72}$.

Observation 4.2 (M_{odd} negative definiteness)**.** *The odd block M_{odd} is negative definite at all tested $\lambda^2 \in [10, 50{,}000]$, with:*

- $\text{tr}(M_{\text{odd}}) \sim -0.92\, L^{2.28}$ *(strongly negative, growing);*
- $\lambda_{\max}(M_{\text{odd}}) \sim -c \cdot L^{-0.35}$ *(near zero, approaching from below);*
- $|\lambda_{\max}|/|\,\text{tr}\,| \sim L^{-2.64}$ *(negativity increasingly robust).*

The maximum eigenvalue is N-converged (stable under basis truncation refinement), confirming it is not a truncation artifact.

5 Anatomy of the Critical Direction

Observation 5.1 (The tightest point)**.** *The maximum eigenvalue of M_{odd} at $\lambda^2 = 1000$ is -1.58×10^{-7}, achieved on the eigenvector*

$$\mathbf{v}_{\text{crit}} \approx -0.54\,|1\rangle + 0.84\,|2\rangle + O(0.07), \tag{10}$$

where $|n\rangle$ denotes the n-th odd basis function $(|n\rangle - |-n\rangle)/\sqrt{2}$. The two-mode concentration exceeds 99% *at all tested λ.*

Observation 5.2 (Ten-figure cancellation)**.** *On the critical direction* (10)*, the Rayleigh quotient of M decomposes as:*

Component	*Rayleigh quotient*
$W_{0,2}$	≈ 0
M_{prime}	$-1.534\,465$
M_{diag}	$+1.564\,967$
M_α	$-0.030\,502$
M_{total}	-1.58×10^{-7}

The archimedean diagonal M_{diag} and the prime sum M_{prime} cancel to ten significant figures, with M_α providing the small negative correction that tips the total below zero.

Observation 5.3 (Variation with λ)**.** *As λ grows:*

λ^2	M_{prime}	M_{diag}	M_α	M_{total}
50	-0.824	$+0.829$	-0.005	-1.84×10^{-7}
200	-1.173	$+1.193$	-0.020	-1.74×10^{-7}
1000	-1.520	$+1.551$	-0.031	-1.61×10^{-7}
5000	-1.838	$+1.871$	-0.034	-1.48×10^{-7}
20000	-2.105	$+2.132$	-0.027	-1.36×10^{-7}

M_{diag} consistently exceeds $|M_{\text{prime}}|$ on this direction, with M_α ensuring the total stays negative. The margin $|M_{\text{total}}| \sim L^{-0.35}$ shrinks but remains negative at all tested points.

Observation 5.4 (The mechanism)**.** *M_{prime} alone has 28–48 positive eigenvalues on the odd block, while M_{diag} has 6–10 positive entries ($\text{wr}_{\text{diag}}(n) > 0$ for small n). The total M_{odd} has* zero *positive eigenvalues. The strongly negative trace of M_{diag} (≈ -34 at $\lambda^2 = 1000$) overwhelms all secondary positive eigenvalues of M_{prime}, while M_{prime}'s single dominant positive mode (at $+34.9$, even parity) is absent from the odd block.*

Standard matrix bounds fail on this problem: Gershgorin gives $+253$ where the actual maximum eigenvalue is -10^{-7} (a factor of 10^8 too loose). Weyl perturbation with $\|M_{\text{prime}}\|_2 \approx 27$ versus $|\lambda_{\max}(M_{\text{diag}} + M_\alpha)| \approx 2.7$ is an order of magnitude short.

6 Connections

6.1 Lorentzian polynomials and Huh–Brändén theory

A symmetric matrix A with signature $(1, d-1)$ is said to have *Lorentzian signature*, after the Huh–Brändén theory [2] where homogeneous polynomials whose Hessians have this signature characterize matroids and M-convex sets.

Our Conjecture 3.1(i) asserts that the Weil matrix M is Lorentzian. This connects RH to the combinatorial Hodge theory that earned Huh the 2022 Fields Medal [1]. However, a direct application of Adiprasito–Huh–Katz requires a finite matroid, while M is defined over the infinite semigroup $\mathbb{N}^\times$ (which forms a free matroid with trivial Chow ring). Any proof of Conjecture 3.1 via this route would require extending Lorentzian theory to Cauchy–Loewner matrices with logarithmically decaying diagonal—a new result in matrix analysis.

6.2 Connes' zeta spectral triples

Connes and Consani [4] construct self-adjoint operators from rank-1 perturbations of the scaling operator on $[\lambda^{-1}, \lambda]$, whose spectra converge to zeta zeros. Their truncation matrix (Lemma 5.1 of [4]) has the form

$$\tau_{ij} = a_i\, \delta_{ij} + \frac{b_i - b_j}{i - j}, \quad i \neq j,$$

which is *exactly* the Cauchy–Loewner structure (8) of our matrix M, verified to relative precision 5×10^{-16} (Observation in Session 59b). The matrices are the same object in different bases (Fourier modes vs. Chebyshev-like functions on $[\lambda^{-1}, \lambda]$).

Connes' approach to Weil positivity uses the Carathéodory–Fejér theorem for Toeplitz matrices. Our analysis shows that M is approximately 15% non-Toeplitz (with the non-Toeplitz part dominated by the logarithmic variation of $\text{wr}_{\text{diag}}(n)$), so the CF criterion does not apply directly. However, the Toeplitz *approximation* of M_{odd} *is* negative definite (maximum eigenvalue -1.03), with $10^7\times$ more margin than M_{odd} itself (-1.6×10^{-7}). The non-Toeplitz correction nearly exactly erases this margin on the critical direction.

6.3 Displacement rank and the explicit formula

The matrix M has displacement rank exactly 2 (verified at 80-digit precision), consistent with the Cauchy–Loewner form (8) having two generator sequences $\{a_n\}$ and $\{B_n\}$. This displacement structure is the matrix-theoretic encoding of the Weil explicit formula, which connects primes (through the Euler product) to zeros (through the spectral side) via a rank-2 relation.

7 Circularity: The Lorentzian Property Is Prime-Specific

Extensive diagnostic testing reveals that the Lorentzian signature of M cannot be proved from the Cauchy–Loewner structure alone.

Observation 7.1 (Cramér model failure)**.** *Replacing real primes by Cramér-model random primes (independent Bernoulli with $P(n \text{ prime}) = 1/\log n$), the odd block M_{odd} has a* positive *maximum eigenvalue at every tested λ: mean $+3.8$ to $+7.7$, with* 0 *out of* 150 *trials producing $M_{\text{odd}} < 0$. The Lorentzian property is* 100% *specific to real primes.*

Observation 7.2 (Every prime is load-bearing)**.** *Removing any single prime p from the sum (even $p = 97$ with weight $\log(97)/\sqrt{97} \approx 0.46$) flips the maximum eigenvalue of M_{odd} from -1.6×10^{-7} to $+0.47$ or more—a perturbation 10^6 times the margin. The property requires the complete prime sum.*

Observation 7.3 (Transition universality)**.** *Adding prime powers cumulatively (sorted by log-height), M_{odd} crosses from positive to negative when* 97–98% *of prime powers are included, regardless of λ. The first ~ 100 prime powers* raise *$\lambda_{\max}(M_{\text{odd}})$ from $+2.7$ (archimedean only) to $+7.0$; the remaining prime powers bring it to -1.6×10^{-7}.*

These diagnostics rule out any purely matrix-theoretic proof of Conjecture 3.1. The Lorentzian property encodes the precise distribution of primes, which is governed by the zeros of ζ on the critical line.

8 A Zero-Detecting Sign Lemma

Although the Lorentzian property cannot be proved from matrix structure alone, it can be *verified* with extreme sensitivity using first-order perturbation theory.

Definition 8.1. *For a nontrivial zero $\rho = \frac{1}{2} + i\gamma$ of the Riemann zeta function, define the* ***perturbation matrix***

$$P_{nm}(\gamma) = -2\,\mathrm{Re}\left[\int_0^L K(n,m,t)\, t\, e^{i\gamma t}\, dt\right], \tag{11}$$

where $K(n,m,t)$ is the prime-sum kernel from (2)–(3)*. This is the first-order eigenvalue response to shifting ρ off the critical line by $\delta > 0$.*

Theorem 8.1 (Sign lemma)**.** *Let $\mathbf{v}$ be the eigenvector of M_{odd} corresponding to its maximum eigenvalue. Then*

$$\mathbf{v}^T P_{\text{odd}}(\gamma)\, \mathbf{v} > 0 \quad \textit{for all } \gamma \geq \gamma_1 = 14.1347\ldots \tag{12}$$

That is, moving any zero off the critical line increases $\lambda_{\max}(M_{\text{odd}})$.

The proof rests on three observations. First, $\mathbf{v}$ is concentrated on the $n = 1, 2$ modes (97%+ of the norm, Observation 5.1). Second, P_{odd} restricted to $\{n = 1, n = 2\}$ is positive definite. Third, the diagonal entries satisfy the closed-form approximation

$$P_{\text{odd}}[n,n] \approx \frac{4}{\gamma^2 - (2\pi n/L)^2}, \tag{13}$$

verified to 0.03% relative precision for $L \geq 7$. Since $\gamma \geq 14.13 > 2\pi \cdot 2/L$ for all $L > 0.89$, both diagonal entries are positive, giving (12).

Remark 8.1 (Critical shift)**.** *The smallest δ that flips $\lambda_{\max}(M_{\text{odd}})$ from negative to positive is $\delta_{\text{crit}} \approx 1.4 \times 10^{-6}$, constant across all tested λ. This is* 2,000 *times more sensitive than the Vinogradov–Korobov zero-free region ($\delta_{\text{VK}} \approx 3.6 \times 10^{-3}$).*

Remark 8.2 (Kernel vanishing)**.** *The diagonal kernel* $K(n,n,t) = 2(L-t)/L \cdot \cos(2\pi nt/L)$ *satisfies* $K(n,n,L) = 0$*. In integration by parts, this kills the* $e^{\delta L}$ *boundary term that would otherwise cause exponential growth of the perturbation with* L*. Consequently, the eigenvalue shift from a zero at height* γ *is* $O(1/\gamma^2)$ independently of L*, confirmed numerically to* 6% *variation across* $L = 3.9$ *to* 11.5*.*

9 Tail Bound and Effective Verification Range

Combining the sign lemma with known zero verification yields a quantitative bound on the unverified zero tail.

Theorem 9.1 (Tail bound)**.** *Let* T *be the height up to which all nontrivial zeros of* ζ *have been verified to lie on the critical line (*$T \approx 3 \times 10^{12}$ *by current computation [10]). Then the total first-order perturbation from all zeros above height* T *satisfies*

$$\sum_{\gamma_k > T} \delta_k \cdot \mathbf{v}^T P_{\mathrm{odd}}(\gamma_k)\, \mathbf{v} \;\le\; \frac{2}{\pi} \cdot \frac{\log T + 1}{T} \;\approx\; 3.1 \times 10^{-12}, \tag{14}$$

where $\delta_k \le \frac{1}{2}$ *is bounded by the Vinogradov–Korobov zero-free region.*

The Schur margin of M_{odd} at the computational verification limit ($\lambda^2 = 287{,}000$, $L = 12.6$) is 2.6×10^{-7}, giving a safety factor of 80,000. The kernel-vanishing property (Remark 8.2) ensures the non-perturbative correction is also L-independent, bounded by $\sim 4 \times 10^{-8}$—still 6 times below the margin.

The margin decays as $\sim 3 \times 10^{-6} L^{-0.97}$ while the tail is constant, crossing at $L \approx 240{,}000$ ($\lambda^2 \approx e^{240{,}000}$). The effective verification range is thus $\lambda^2 < e^{240{,}000}$—far beyond any physical scale but finite.

10 Geometric Structure

The matrix M sits naturally in a Hodge-theoretic framework.

Observation 10.1 (Hodge decomposition)**.** *The null space of* $W_{0,2}$ *decomposes by parity into two blocks of dimension* $\lfloor N/2 \rfloor$ *and* $\lceil N/2 \rceil$*. The restriction of* M *to* each *parity block has zero positive eigenvalues at all tested* λ*. This is the discrete analogue of the Hodge–Riemann bilinear relations on* $H^{1,0} \oplus H^{0,1}$*.*

The correspondence with classical Kähler geometry is:

Classical surface	Connes–Weil matrix
$H^0 \oplus H^2$	range($W_{0,2}$)
H^1 (primitive)	null($W_{0,2}$)
Hodge star $*$	Parity involution $n \mapsto -n$
Lefschetz $L : H^0 \to H^2$	$W_{0,2}$-induced map $\mathbf{u} \mapsto \mathbf{w}$
Hodge–Riemann: $Q < 0$ on H^1_{prim}	$M \le 0$ on null($W_{0,2}$)

The second eigenvalue of M is continuous in λ, always negative, and approaches zero from below as $\lambda \to \infty$ with rate $\sim L^{-0.35}$. The Lorentzian cone (at most one positive eigenvalue) is closed, and M never exits it in the verified range. However, the asymptotic approach to zero prevents extending this argument to all λ without additional input.

11 The Zero-Encoding Constant

The sign lemma's total sensitivity is controlled by a single constant.

Definition 11.1 (The locking constant)**.** *Define*

$$\mathcal{Z} = \frac{\xi''(\frac{1}{2})}{\xi(\frac{1}{2})} = 0.046209986230837941\ldots \tag{15}$$

This constant connects three worlds: the archimedean place (through π and Γ), the prime distribution (through ζ), and the zeros (through the Hadamard sum). It decomposes as

$$\mathcal{Z} = \underbrace{\left(-8 + \frac{\pi^2}{4} + 2G\right)}_{A = -3.7007\ldots} + \underbrace{\left(\frac{\zeta''}{\zeta} - \left(\frac{\zeta'}{\zeta}\right)^2\right)\Bigg|_{s=1/2}}_{P = +3.7469\ldots} \tag{16}$$

where $G = 0.9160\ldots$ is Catalan's constant. The archimedean part A involves only π and G; the prime part P requires ζ. Their 80:1 cancellation ($|A|/\mathcal{Z} \approx 80.08$) leaves the small positive residual $\mathcal{Z}$.

Observation 11.1 (Locking identity)**.** *From the Hadamard product, each zero pair contributes*

$$f(\delta_k, \gamma_k) = \frac{2(\gamma_k^2 - \delta_k^2)}{(\delta_k^2 + \gamma_k^2)^2} \leq \frac{2}{\gamma_k^2}, \tag{17}$$

with equality if and only if $\delta_k = 0$ (on-line). The sum is fixed:

$$\mathcal{Z} = \sum_{\text{pairs}} f(\delta_k, \gamma_k) \leq \sum_{\text{pairs}} \frac{2}{\gamma_k^2}. \tag{18}$$

Equality holds in (18) *if and only if every $\delta_k = 0$, i.e., if and only if RH is true.*

The constant $\mathcal{Z}$ is computable to arbitrary precision without reference to the zeros, via the Dirichlet eta function $\eta(s) = \sum(-1)^{n+1}n^{-s}$, which converges at $s = \frac{1}{2}$. The relation $\zeta(s) = \eta(s)/(1 - 2^{1-s})$ bypasses the natural boundary of the prime zeta function at $\mathrm{Re}(s) = 1$.

Remark 11.1 (Identification)**.** *The locking constant is twice the Voros superzeta function [11] at $s = 2$:*

$$\mathcal{Z} = 2\,\mathcal{Z}_1(2), \qquad \mathcal{Z}_1(s) = \sum_{\gamma_k > 0} \frac{1}{\gamma_k^s}. \tag{19}$$

It satisfies $\mathcal{Z} \approx 2 + \gamma_{\mathrm{E}} - \log(4\pi)$ to 0.04%, where the right side is the Hadamard constant $\sum_\rho 1/(\rho(1-\rho))$. The 0.04% discrepancy (9.28×10^{-6}) arises from the higher moment $\sum 1/(4\gamma^2(\frac{1}{4}+\gamma^2))$. The decomposition (16)*, the* 80:1 *cancellation ratio, the locking identity* (18)*, and the connection to the Cauchy–Loewner matrix framework appear to be new.*

Remark 11.2 (Hilbert–Pólya operator)**.** *The Taylor expansion $\xi(\frac{1}{2} + i\sqrt{w})/\xi(\frac{1}{2}) = \sum c_k w^k$ defines a polynomial whose roots are $w_k = \gamma_k^2$. Its companion matrix has eigenvalues γ_k^2 and is constructed entirely from the derivatives $\xi^{(2k)}(\frac{1}{2})/\xi(\frac{1}{2})$, computable via the eta function without reference to the zeros. With $K = 6$ moments, the first six zeros are recovered to full floating-point precision. This provides an explicit (though ad hoc) Hilbert–Pólya-type operator.*

Remark 11.3 (Physics parallel: misaligned supersymmetry)**.** *The* 80:1 *cancellation between archimedean and arithmetic contributions is structurally parallel to Bose–Fermi cancellations in non-supersymmetric gauge theories, where modular invariance of the partition function forces the vacuum energy to be non-negative despite massive cancellations between bosonic and fermionic state counts [8]. In both settings:*

- *Two structurally distinct sectors (geometric/archimedean vs. arithmetic/prime, or bosonic vs. fermionic) contribute terms of opposite sign;*
- *A symmetry constraint (functional equation vs. modular invariance) forces near-cancellation;*
- *The residual is non-negative, with its positivity encoding deep structural information.*

The spectral action program of Connes and Chamseddine [9] connects these two worlds through noncommutative geometry, but no one has yet closed the loop from the physics of modular invariance to a proof of Weil positivity.

Remark 11.4 (Open computation)**.** *Connes and Consani's spectral triples [4] construct operators H_N whose eigenvalues approximate the zeta zeros with extraordinary precision. The spectral invariant $\mathrm{tr}(H_N^{-2})$ should converge to $\mathcal{Z}/2$ as $N \to \infty$, providing $\mathcal{Z}$ with a spectral-geometric interpretation. This computation has not been performed.*

Remark 11.5 (Turán inequalities and the Jensen–Pólya program)**.** *The Taylor coefficients $c_k = (-1)^k \xi^{(2k)}(\frac{1}{2})/(2k)!$ of $\xi(\frac{1}{2}+it)/\xi(\frac{1}{2}) = \sum c_k t^{2k}$ satisfy the Turán inequalities $c_k^2 \geq c_{k-1}c_{k+1}$ for $k = 1, \ldots, 15$ (our computation), with ratios $c_k^2/(c_{k-1}c_{k+1})$ decreasing from 2.15 to 1.09. These inequalities are necessary and sufficient for $\xi(\frac{1}{2} + it)$ to belong to the Laguerre–Pólya class of entire functions with only real zeros, which is equivalent to RH.*

*Griffin, Ono, Rolen, and Zagier [**?**] proved unconditionally that the Jensen polynomials of ξ are hyperbolic (all real zeros) for all degrees $d \geq 1$ at sufficiently large shifts, and for all shifts when $d \leq 8$ (extended to $d \leq 2 \times 10^{17}$ by Chasse). For large shifts, the Jensen polynomials converge to Hermite polynomials—a manifestation of GUE universality.*

The gap—hyperbolicity at small shifts for large degrees—has the same structure as our sign lemma wall: the asymptotic regime is handled, finite cases need checking, and the checking never reaches all cases. The Turán ratios approaching 1 from above correspond to our Schur margin approaching 0 from below.

The structural parallel with misaligned supersymmetry (Remark 11.3) suggests a path: in non-supersymmetric string theory, modular invariance forces log-concavity of partition function coefficients at all *levels simultaneously [**?**]. The functional equation $\xi(s) = \xi(1-s)$ is the number-theoretic analogue of modular invariance. A proof that the functional equation forces uniform Turán inequalities would close the Jensen–Pólya program and prove RH.*

12 Conclusions

We have formulated the Lorentzian Weil Matrix Conjecture (Conjecture 3.1), proved it implies RH (Theorem 3.1), and developed a sign lemma (Theorem 8.1) that detects off-line zeros with sensitivity $\delta_{\mathrm{crit}} \approx 1.4 \times 10^{-6}$—three orders of magnitude beyond the Vinogradov–Korobov zero-free region.

The kernel-vanishing property $K(L) = 0$ eliminates exponential growth of the zero-tail perturbation, yielding an L-independent bound (Remark 8.2). Combined with verified zeros up to height 3×10^{12} and computational verification of $Q_W \geq 0$ to $\lambda^2 = 287{,}000$, the effective verification range extends to $\lambda^2 < e^{240{,}000}$ with a safety factor of 80,000 (Theorem 9.1).

The locking constant $\mathcal{Z} = \xi''(\frac{1}{2})/\xi(\frac{1}{2})$ encodes the total sensitivity of the Lorentzian property to zero displacement. It arises from an 80:1 cancellation between the archimedean contribution ($A = -8 + \pi^2/4 + 2G$, involving only π and Catalan's constant) and the prime contribution ($P \approx 3.747$, involving ζ at the critical point). The locking identity (18) reformulates RH as the equality $\mathcal{Z} = \sum 2/\gamma_k^2$—each zero contributes maximally when on the critical line. This identity is verifiable to arbitrary precision: the left side through the eta function, the right side through the computed zeros.

Extensive diagnostics establish that the Lorentzian property cannot be proved from the Cauchy–Loewner structure alone (the Cramér model gives 0/150 negative-definite trials), nor from the geometric Hodge framework (which maps cleanly but hits the same quantitative wall), nor from any convergent prime sum (the natural boundary of the prime zeta function at $\mathrm{Re}(s) = 1$ prevents continuation to the critical point). The Lorentzian property encodes the specific distribution of primes enforced by the zeros of ζ, and any proof must engage with this arithmetic content.

The remaining gap—the margin decays as $L^{-0.97}$ while the tail is a positive constant—is structural: it requires either a positive lower bound on the margin (which would be stronger than RH) or a fundamentally new argument that connects the archimedean–prime cancellation to the zero distribution without passing through the natural boundary.

The structural parallel with misaligned supersymmetry (Remark 11.3) suggests a possible path: modular invariance of partition functions forces Bose–Fermi cancellations to leave non-negative residuals in non-supersymmetric gauge theories. The functional equation of ζ is the number-theoretic analogue of modular invariance. A proof that this analogy can be made rigorous—that the functional equation forces the locking identity (18) to hold with equality—would constitute a proof of RH. The uncomputed spectral invariant $\mathrm{tr}(H_N^{-2})$ for Connes' operators is a concrete first step.

Acknowledgments

This work was conducted over 67 computational sessions using the Connes–Consani–Moscovici framework, with literature review assisted by automated research agents. All numerical results are reproducible from the Python scripts in the project repository.

References

[1] K. Adiprasito, J. Huh, and E. Katz, *Hodge theory for combinatorial geometries*, Ann. of Math. **188** (2018), 381–452. arXiv:1511.02888.

[2] P. Brändén and J. Huh, *Lorentzian polynomials*, Ann. of Math. **192** (2020), 821–891.

[3] A. Connes and C. Consani, *Weil positivity and trace formula, the archimedean place*, Selecta Math. (2021). arXiv:2006.13771.

[4] A. Connes and C. Consani, *Zeta spectral triples*, preprint (2025). arXiv:2511.22755.

[5] A. Connes, *The Riemann Hypothesis: past, present, and a letter through time*, preprint (2026). arXiv:2602.04022.

[6] I. Markovsky, *Sum-of-exponentials via Hankel low-rank approximation*, preprint (2014).

[7] M. Morishita, *On a relation between Deninger's foliated dynamical systems and Connes–Consani's adelic spaces*, Münster J. Math. (2025). arXiv:2508.15971.

[8] G. Başar, A. Cherman, and D. McGady, *Bose–Fermi cancellations without supersymmetry*, Phys. Rev. D **99** (2019), 065014. arXiv:1812.04642.

[9] K. R. Dienes, *Modular invariance, finiteness, and misaligned supersymmetry*, Nucl. Phys. B **429** (1994), 533–588. arXiv:hep-th/9402006.

[10] M. Griffin, K. Ono, L. Rolen, and D. Zagier, *Jensen polynomials for the Riemann zeta function and other sequences*, Proc. Natl. Acad. Sci. USA **116** (2019), 11103–11110.

[11] A. Connes and A. Chamseddine, *The spectral action principle*, Comm. Math. Phys. **186** (1997), 731–750. arXiv:hep-th/9606001.

[12] D. Platt and T. Trudgian, *The Riemann hypothesis is true up to* 3×10^{12}, Bull. London Math. Soc. **53** (2021), 792–797.

[13] A. Voros, *Zeta functions for the Riemann zeros*, Ann. Inst. Fourier **53** (2003), 665–699.

[14] R. Roy and T. Kailath, *ESPRIT—estimation of signal parameters via rotational invariance techniques*, IEEE Trans. ASSP **37** (1989), 984–995.